Idiosyncratic Risk

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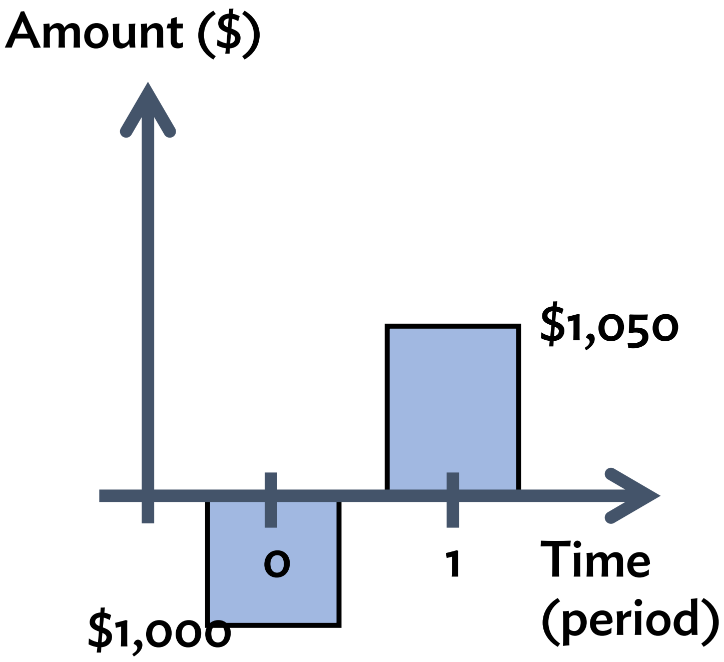
Let’s bet $1,000 on a coin flip

- We have a fair coin
  - Fair meaning: 50-50% chance of “heads” and “tails”
- And we have $1,000 to bet (invest) on a single coin toss
- **Heads:** Pays $1.20 for each $1 invested (20% return)
  - Total payoff is $1,200 ($1.20 × $1,000)
- **Tails:** Pays $0.90 for each $1 invested (-10% return)
  - Total payoff is $900 ($0.90 × $1,000)
- Expected return is 5%
  - Because, $1,200 × 50% + $900 × 50% = $1,050
  - And, $1,050 / $1,000 = 1.05 ~ 5% return

$\$1,000 \xrightarrow{H} \$1,200$

$\$1,000 \xrightarrow{T} \$900$

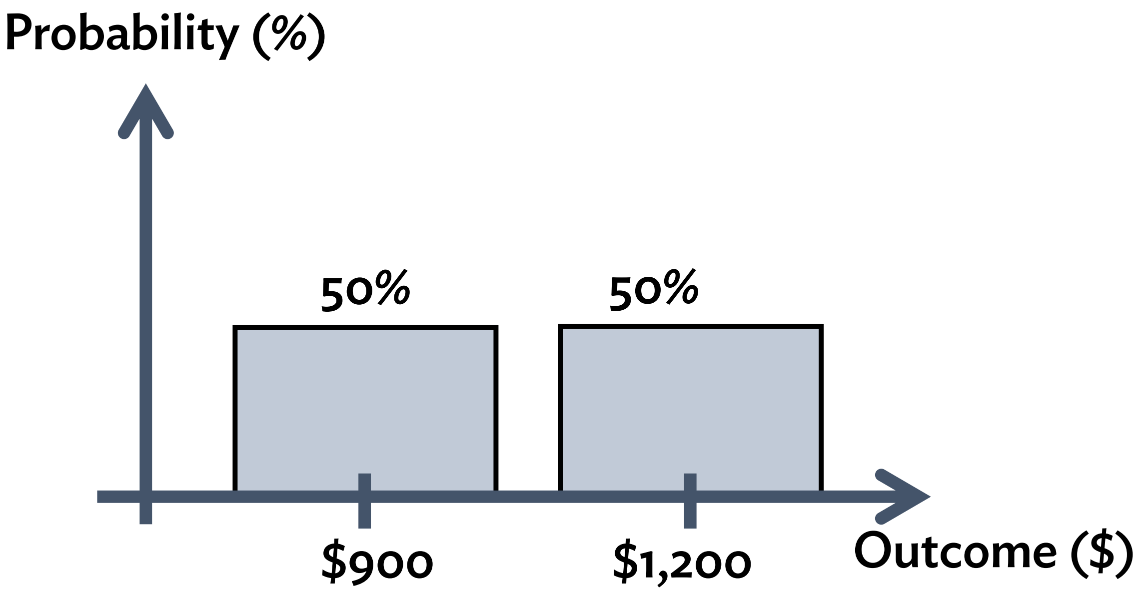
Cash flow from a coin flip



- Picture shows average (expected) cash flows
  - Cannot tell risk from picture that coin flip never pays $1,050
  - Somewhat misleading illustration of financial outcome of coin flip

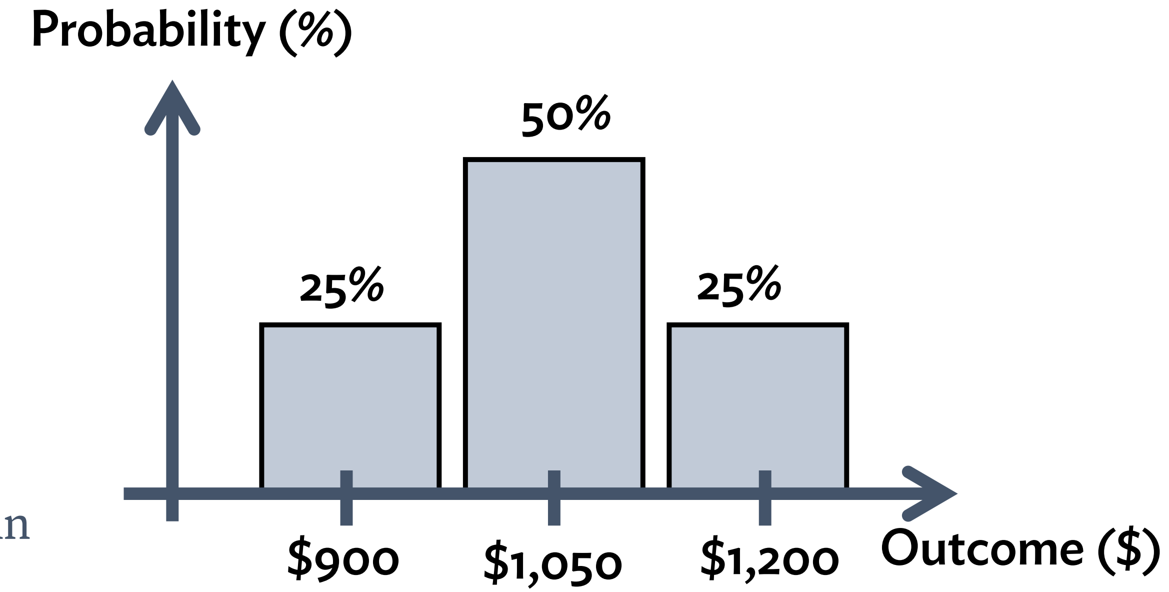
Histogram of coin flip outcomes

- A histogram is another way to show distribution of outcomes



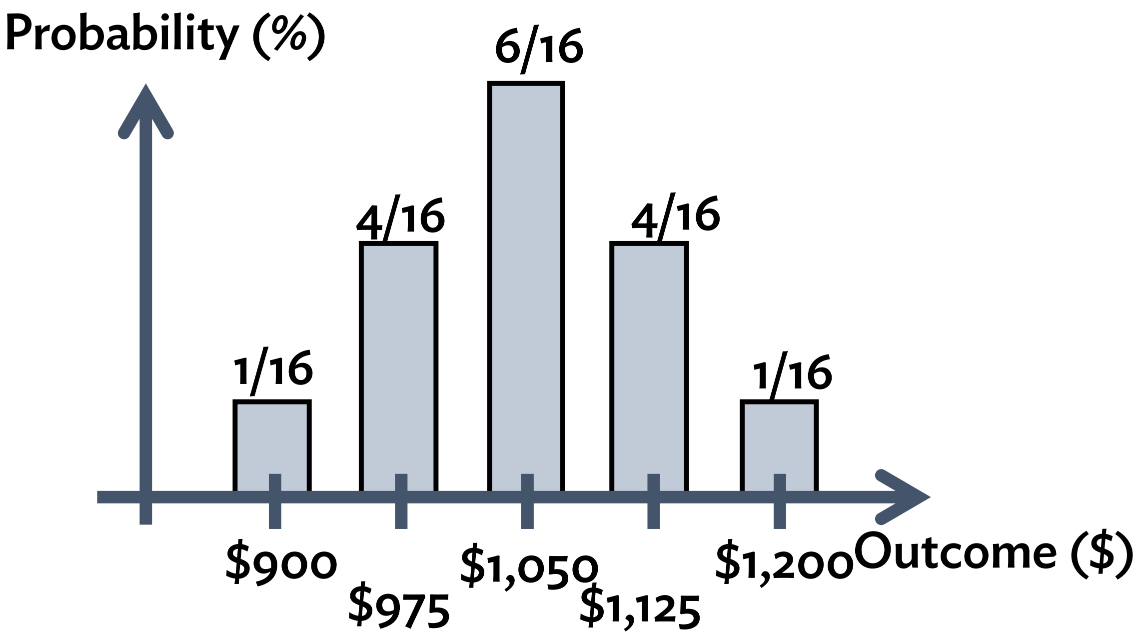
Let's get a second coin

- Now flip two coins at the same time
  - Same payout as before: Heads +20%, Tails -10%
  - Invest $500 in each coin
- Possible outcomes:
  - **HH** (25%): $1,200
  - **HT/TH** (50%): $1,050
  - **TT** (25%): $900
- Expected (average) payout is still $1,050
  - Expected (average) return is still 5%
- But now outcomes are more concentrated in middle and are less risky
- This is diversification



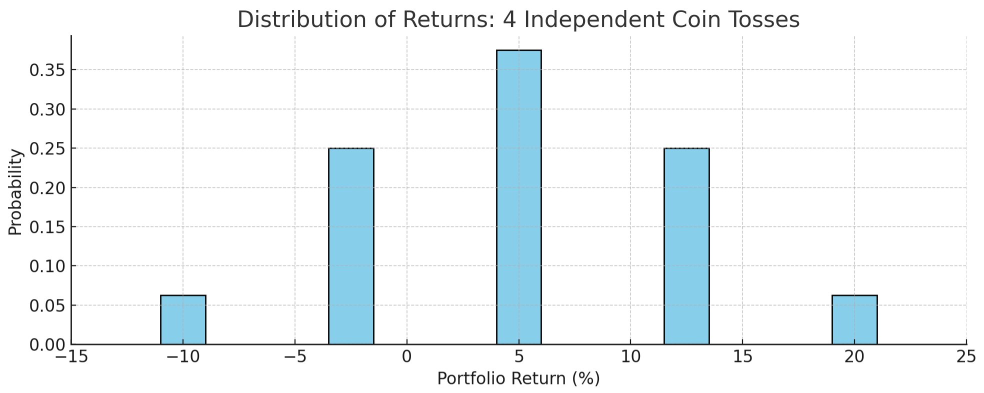
Let's try four coins

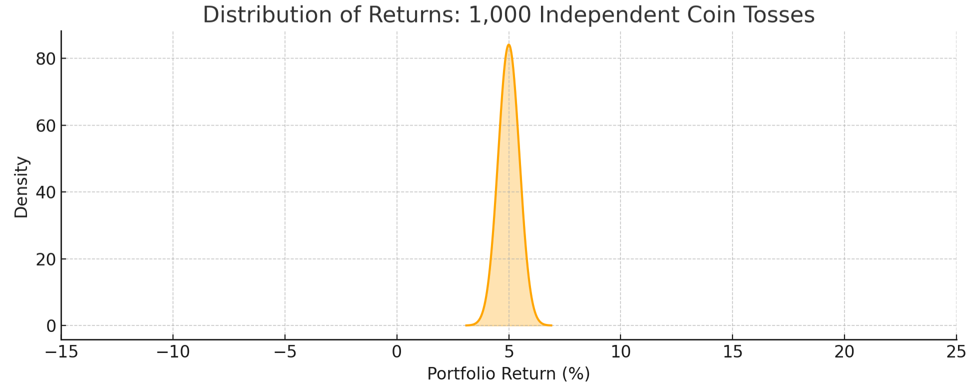
- Now flip four coins with the same payouts
  - Invest $250 in each coin
- five possible outcomes
  - Zero to four heads
- Most likely, we get about two heads and two tails
  - Outcomes are even more clustered in middle
  - Pretty unlikely to get all heads (or all tails)



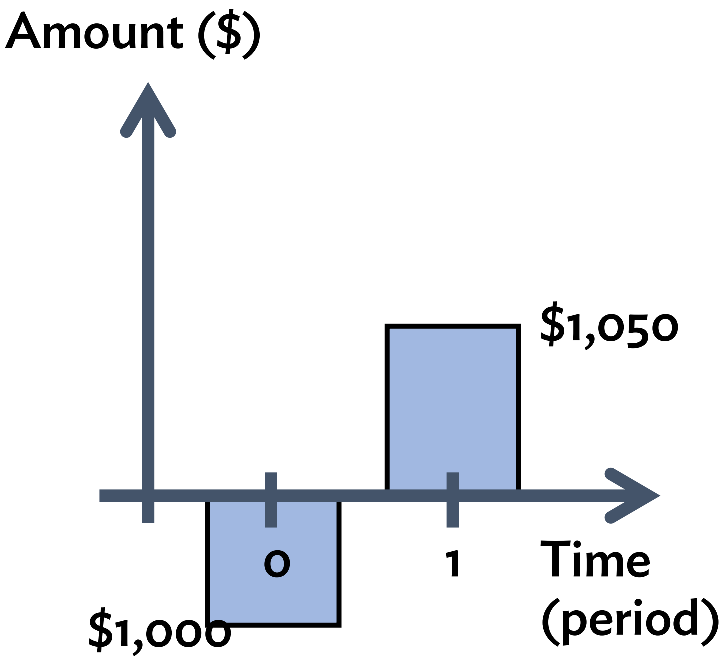
Distribution of returns with 1,000 coins

- Graphs show outcomes with 4 and 1,000 coins
  - Expected return remains 5%
- Actual return increasingly concentrated at 5% (total payuot close to $1,050)
- With many coins, total payout becomes essentially risk-free
  - Each individual coin toss is risky
  - But collectively, across all the coins, idiosyncratic risk disappears
- This is full diversification





Cash flow from 1,000 coin flips



- Graph shows average (expected) cash flows
  - With 1,000 coin flips, payout at period 1 will be close to $1,050 as shown in graph
  - Thus, graph is a fine for illustrating the total payoff of 1,000 coin flips

Diversification

- By dividing investment across more coins, we reduce overall risk, but keep the same expected (average) return
  - In example, expected (average) return remains 5%
  - Middle outcomes become increasingly likely
  - Extreme outcomes become increasingly unlikely
- Investors like expected return and dislike risk, so this is good news
  - Investor would want to diversify and divide its investments across as many "coins" as possible to reduce risk
- With a diversified portfolio of idiosyncratic risks, total payout is (pretty much exactly) the expected return of all the investments
  - Can treat portfolio as delivering this return as a risk-free cash flow

Valuing idiosyncratic risk

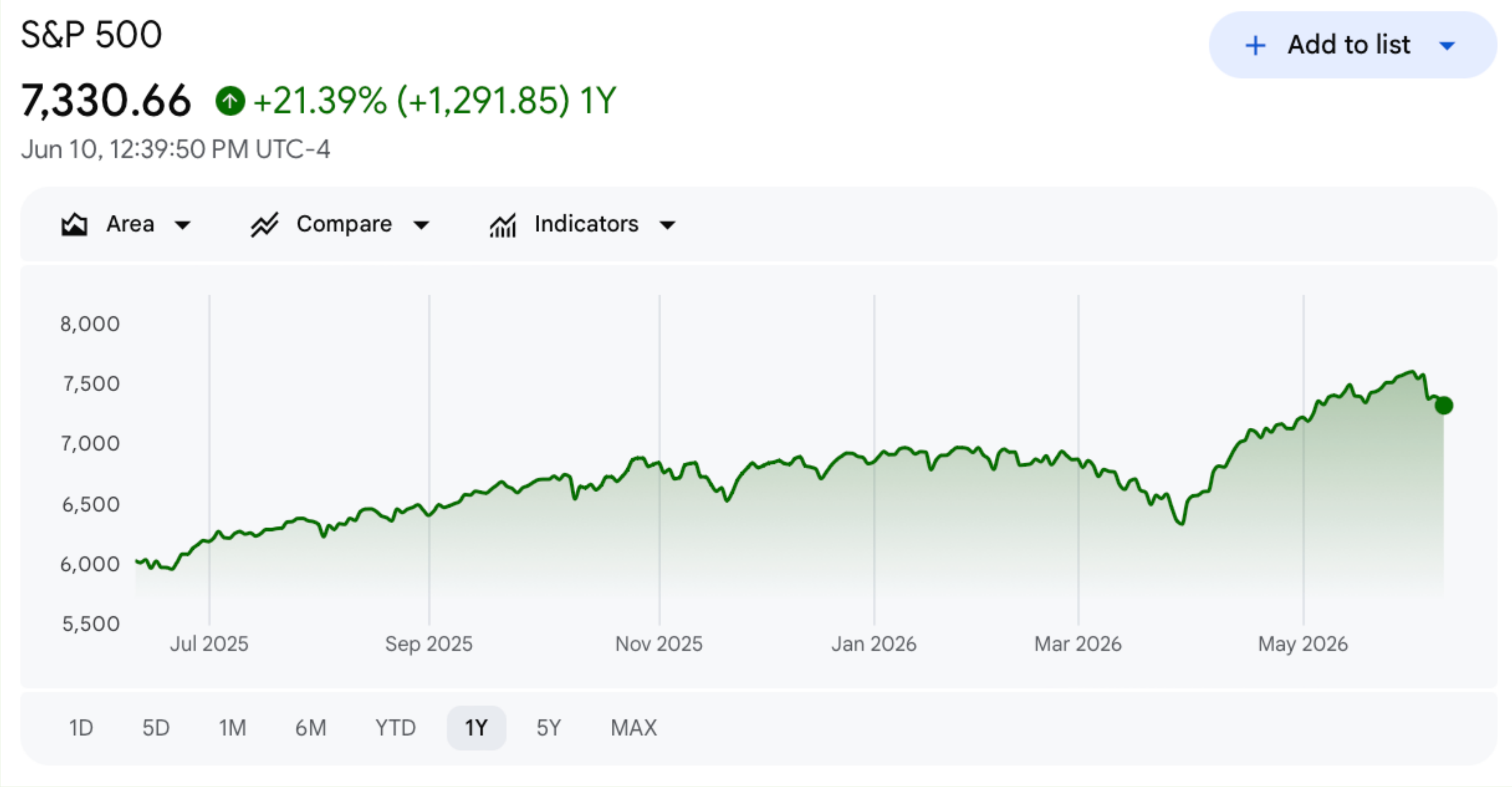
- For example, consider adding one more coin flip to a diversified portfolio
    - Each individual coin flip is risky, it pays either \$1.20 or \$0.90 with 50%–50% chance (\$1.05 expected payout)
- When included in a diversified portfolio, the investor values the coin's payoff as a risk-free cash flow of \$1.05
    - Thus, appropriate discount rate is the risk-free rate
    - Present value is: PV = \$1.05 / (1+ risk-free rate)
- This is very counter-intuitive

Systematic Risk

Systematic risk

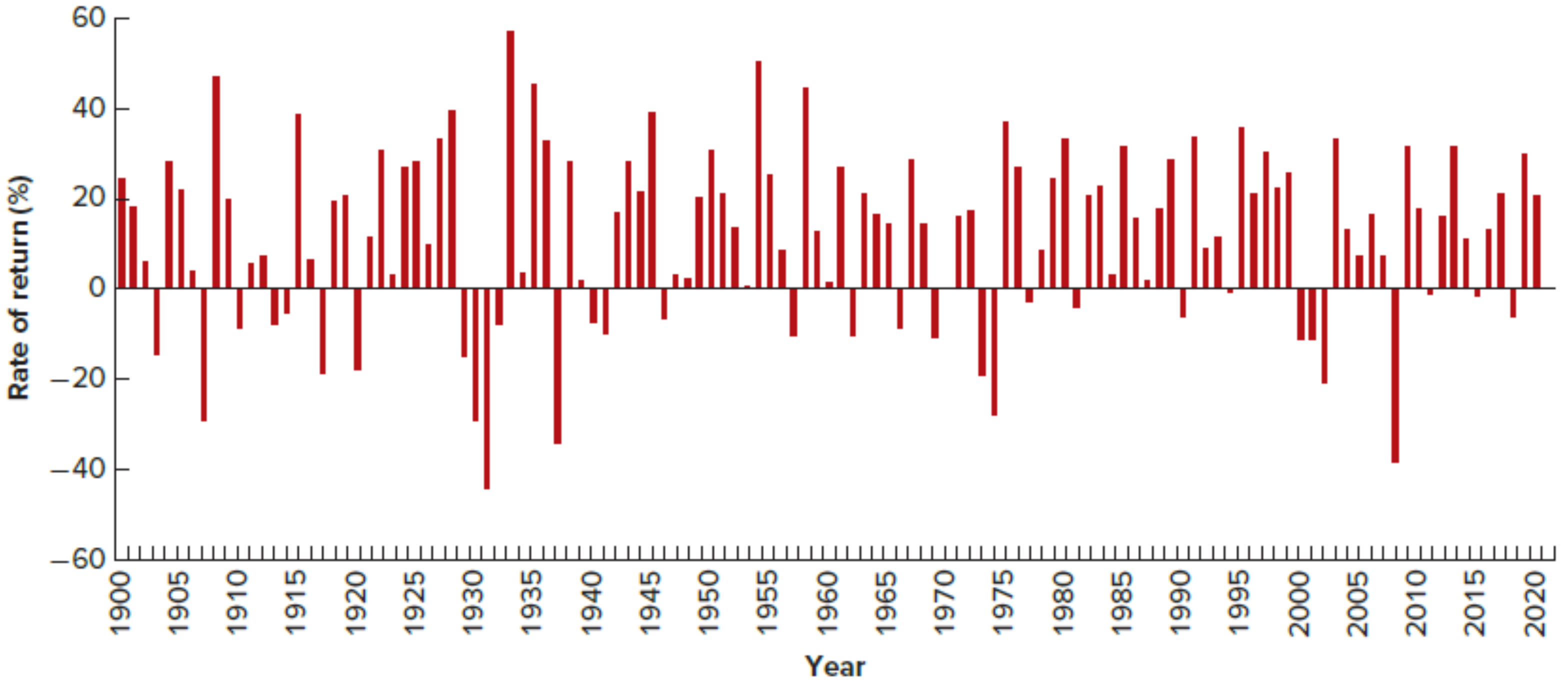
- Instead of investing in coins let's invest in the US stock market
  - US stock market has about 4,000 companies with listed shares
  - A diversified investor in the market owns shares in a broad selection of these companies
- A share's financial return is risky, and depends on the company's performance
  - Generally, when company does well, share price increases, and investors earn a return
- How is the return on a diversified portfolio of shares similar to (and different from) investing in a number of coin tosses?
  - Is idiosyncratic risk diversified away?
  - Is diversified return risk free?

S&P 500 return over past 12 months



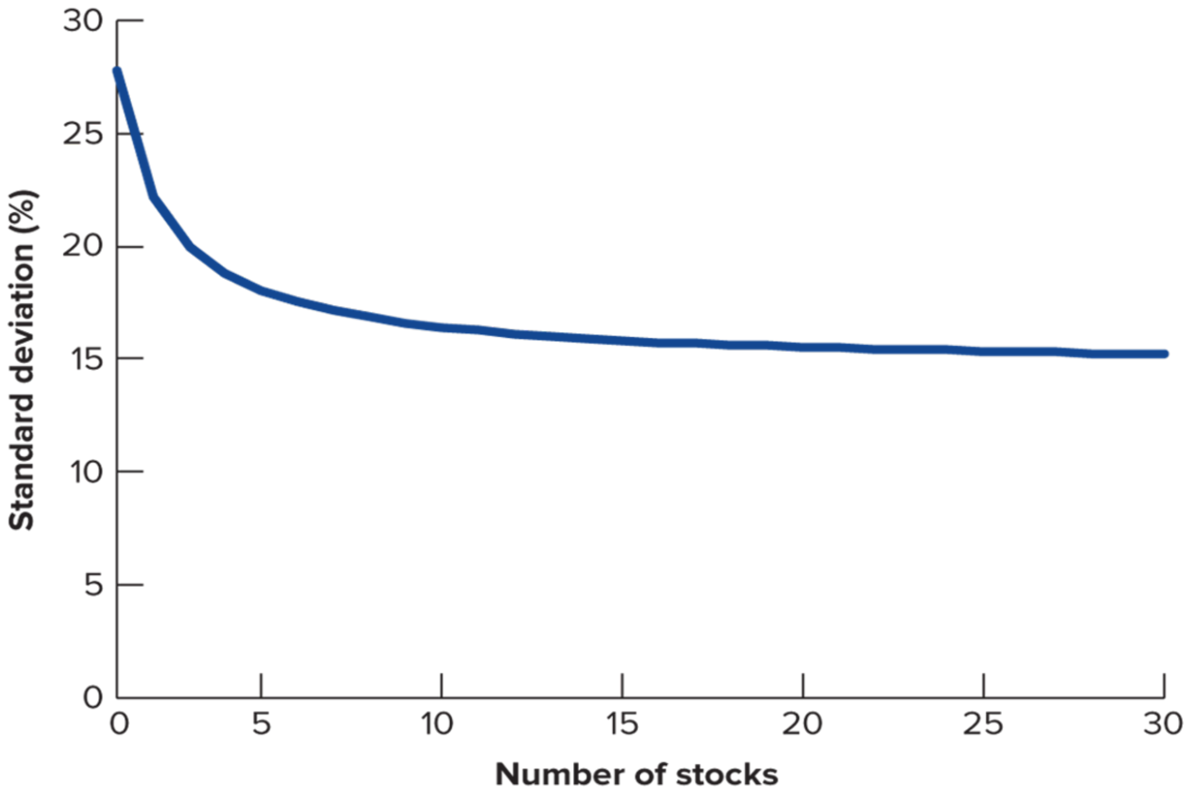
- https://www.google.com/finance/beta/quote/.INX:INDEXSP?window=1Y

Historical US annual market return



- Figure shows annual returns on US stocks from 1900–2020
  - Note swings in 1929–30, 1973, 2001, 2008

Risk and diversification



- Portfolio risk declines with more stocks, but it does not disappear entirely

Systematic risk of stock market return

- A diversified stock portfolio diversifies away the idiosyncratic risk of individual shares
  - Company A may be lucky, Company B may be unlucky, this risk is diversified away
- However, there remain market-wide risks that are not diversified away
  - Due to broad economy-wide shocks
  - A diversified portfolio is still exposed to these risks
  - This remaining risk is systematic risk
- Different investments have different amounts of systematic and idiosyncratic risks
  - Some investments have more systematic risk (*e.g.*, SP500 ETF)
  - Others have more idiosyncratic risk (*e.g.*, coin flip)

Measuring Systematic Risk: Beta

Beta and systematic risk

- The measure of systematic risk is called beta (Greek: β)
- Beta measures how much the return of an investment or cash flow changes with the overall stock market return
  - A beta above 1.0 means the cash flow moves more than the market (more systematic risk)
  - A beta below 1.0 means the cash flow moves less than the market (less systematic risk)
  - A beta of 1.0 means it moves the same as the market (market's own beta is 1.0)
- **High-beta:** Companies and projects that thrive in booms and suffer in downturns
  - Typically, firms in cyclical industries, luxury goods, and discretionary consumer products
- **Low-beta:** Companies and projects in stable industries like utilities, healthcare, or basic consumer staples
  - Stable cash flows and returns that are unaffected by market swings
  - Also, investments with purely idiosyncratic risks that are unrelated to the market, *e.g.*, technological innovation

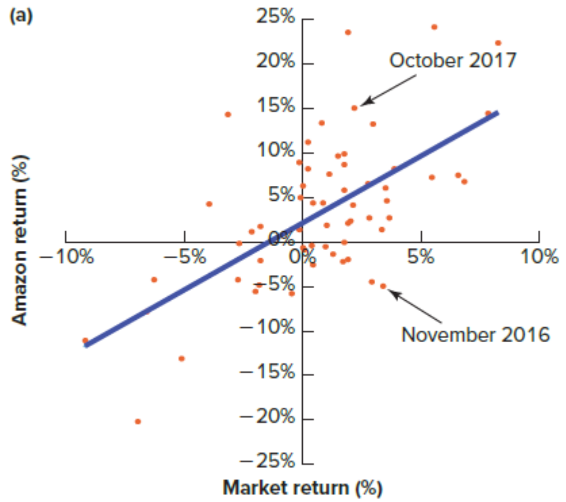
Definition and estimation of Beta

- Mathematical definition of beta:

$$\beta_i = \frac{COV(r_i, r_m)}{VAR(r_m)}$$

- r_i is investment's return
- r_m is return on stock market
- COV and VAR are statistical covariance and variance

- Beta is estimated by statistical regression analysis:



- Luckily, financial data providers estimate and publish betas
  - We don't need to do the statistical regression analysis ourselves

Amazon.com Inc

Added

$239.20 +42.94% (+71.86) 5Y

Jun 10, 12:41:23 PM UTC-4 · USD

Area | Compare | Indicators | Key moments

1D 5D 1M 6M YTD 1Y 5Y MAX

Overview | Earnings | Financials | Holdings New

Amazon declined approximately 8% over the past month, pressured by a massive $200 billion capital exp...

Open	$243.61	Volume	16.71M	Beta	1.45
High	$250.43	P/E ratio	28.61	Shares outstanding	10.76B
Low	$237.00	52-wk high	$278.56	No. of employees	2M
Mkt. cap	2.58T	52-wk low	$196.00		
Avg. vol.	44.07M	EPS	$8.37		

Apple Inc

Added

$292.71 +129.85% (+165.36) 5Y

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Area · Compare · Indicators · Key moments

1D 5D 1M 6M YTD 1Y **5Y** MAX

Overview | Earnings | Financials | Holdings New

Apple advanced approximately 14% over the past month, fueled by record fiscal Q2 results and a massive...

Open	$290.74	Dividend	0.37%	EPS	$8.27
High	$300.75	Quarterly dividend	$0.27	Beta	1.09
Low	$287.38	Ex-dividend date	May 11, 2026	Shares outstanding	14.69B
Mkt. cap	4.30T	P/E ratio	35.39	No. of employees	166K
Avg. vol.	51.52M	52-wk high	$317.40		
Volume	20.98M	52-wk low	$195.07		

NVIDIA Corp

+ Add to list

$203.01 +1,038.56% (+185.18) 5Y

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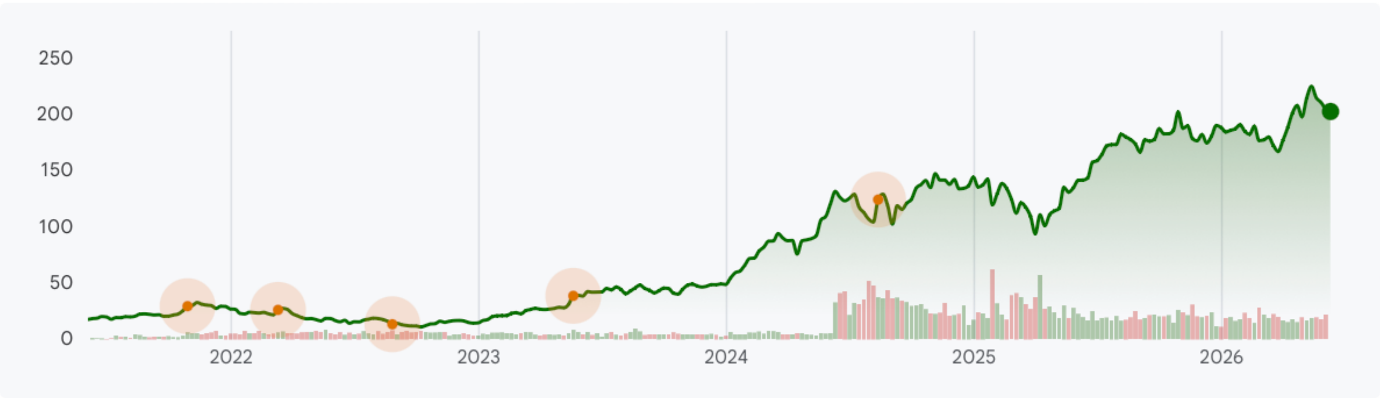


1D 5D 1M 6M YTD 1Y 5Y MAX

Overview | Earnings | Financials | Holdings New

NVIDIA remains on a bullish trajectory despite recent 3.4% dips driven by sector-wide interest rate cautio...

Open	$204.43	Dividend	0.49%	EPS	$6.53
High	$207.22	Quarterly dividend	$0.25	Beta	2.22
Low	$201.61	Ex-dividend date	Jun 4, 2026	Shares outstanding	24.20B
Mkt. cap	4.93T	P/E ratio	31.19	No. of employees	42K
Avg. vol.	170.97M	52-wk high	$236.54		
Volume	77.47M	52-wk low	$140.86		

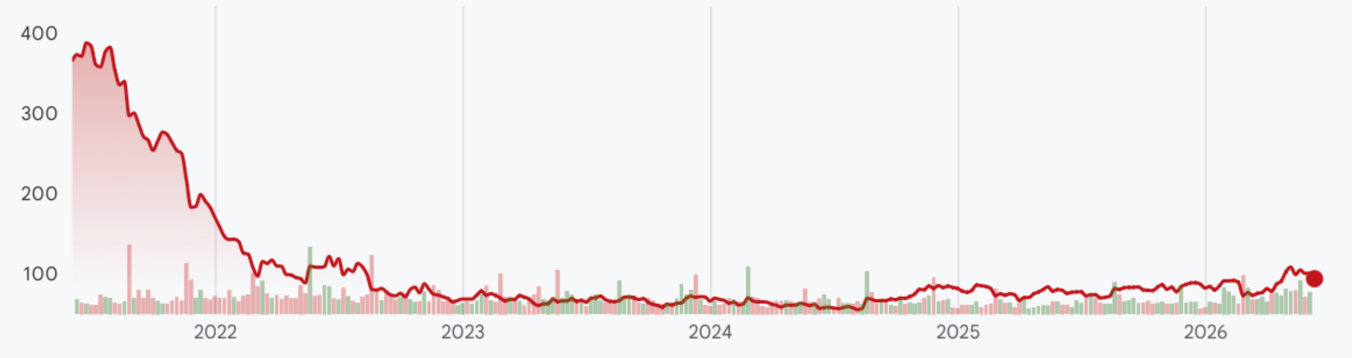
Zoom Communications Inc

+ Add to list

$94.83 -74.12% (-271.57) 5Y

Jun 10, 12:30:29 PM UTC-4 · USD

Area Compare Indicators



1D 5D 1M 6M YTD 1Y 5Y MAX

Overview Earnings Financials Holdings New

Zoom Communications declined approximately 13% recently due to a short-term sell-off despite a positiv...

Open	$95.50	Volume	1.14M	Beta	1.01
High	$96.85	P/E ratio	13.92	Shares outstanding	264.65M
Low	$94.51	52-wk high	$114.74	No. of employees	7K
Mkt. cap	27.80B	52-wk low	$69.15		
Avg. vol.	5.32M	EPS	$6.81		

Consolidated Edison Inc

+ Add to list

$107.81 +39.60% (+30.58) 5Y

Jun 10, 12:44:01 PM UTC-4 · USD

Area · Compare · Indicators · Key moments

1D · 5D · 1M · 6M · YTD · 1Y · 5Y · MAX

Overview · Earnings · Financials · Holdings New

Consolidated Edison fell recently as investors reacted to first-quarter earnings misses and defensive sect...

Open	$109.01	Dividend	3.30%	EPS	$5.94
High	$109.02	Quarterly dividend	$0.89	Beta	0.27
Low	$106.70	Ex-dividend date	May 13, 2026	Shares outstanding	368.53M
Mkt. cap	39.70B	P/E ratio	18.14	No. of employees	15K
Avg. vol.	2.55M	52-wk high	$116.23		
Volume	640.78K	52-wk low	$94.96		

Selected industry betas

Industry Name	Beta		Industry Name	Beta
Retail (Building Supply)	1.79		Insurance (Life)	0.73
Software (Internet)	1.69		Food Wholesalers	0.72
Auto & Truck	1.62		Utility (Water)	0.68
Semiconductor	1.49		Rubber& Tires	0.65
Semiconductor Equip	1.48		Publishing & Newspapers	0.64
Homebuilding	1.43		Beverage (Alcoholic)	0.61
Shoe	1.42		Insurance (Prop/Cas.)	0.61
Building Materials	1.36		Retail (Grocery and Food)	0.58
Retail (Automotive)	1.35		Shipbuilding & Marine	0.58
Advertising	1.34		Beverage (Soft)	0.57
Recreation	1.33		Asset Management	0.57
Construction Supplies	1.29		Power	0.54
Electrical Equipment	1.27		Reinsurance	0.54
Drugs (Biotechnology)	1.25		Banks (Regional)	0.52

- **Source:** https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/Betas.html

Capital Asset Pricing Model (CAPM)

Capital Asset Pricing Model (CAPM)

- CAPM gives the expected returns of stocks trading in the market, depending on their systematic risk (beta)
- CAPM is a financial equilibrium model derived under a number of assumptions:
  - Such as: efficient markets, market in equilibrium, no-arbitrage, rational investors with diversified portfolios, linear stochastic discount factor (SDF)
- CAPM is widely used in practice for finding expected returns
  - Even if CAPM is not exactly true, it is reliable and robust

Capital Asset Pricing Model (CAPM)

- CAPM says that expected return for security *i* with β_i of systematic risk is:

$$\boxed{r_i = r_f + \beta_i \times r_{mf}}$$

- r_i is security *i*'s expected return
- r_f is risk-free rate, i.e., the interest rate
- r_{mf} is "market-risk-premium"
  - Typically, considered to be around 5.0% to 7.0% annually
- β_i is systematic risk

CAPM and present value (PV)

PV calculation now has two steps:

- Determine beta and use CAPM to find expected return
  - Cash flows with more systematic risk have higher expected return

$$r_i = r_f + \beta_i \times r_{mf}$$

- Discount future expected cash flows to calculate PV
  - Discount rate is expected return from CAPM (this is opportunity cost of capital)

$$PV_i = \sum_{t=1}^{T} \frac{E[CF_t]}{(1+r_i)^t}$$

CAPM example

- **Example:** What is present value (PV) of the future expected cash flows (E[CF]): \$10, \$12, \$5, and \$10 (arriving in periods 1, 2, 3, and 4)?
- **Assumptions:**
  - Beta is 1.2
  - Risk-free rate is 4.5%
  - Market risk premium is 6.0%

CAPM and PV in Excel

	A	B	C	D	E	F	G	H
1								
2		**Period**	**0**	**1**	**2**	**3**	**4**	
3		Expected Cash flow (E[CF])	$0.00	$10.00	$12.00	$5.00	$10.00	
4		Discounted CF	$0.00	$8.95	$9.62	$3.59	$6.42	=G3/(1+C12)^G2
5		**Present Value (PV)**	**$28.58**	=SUM(C4:G4)				
6								
7		**Discount rate calculation**						
8		Risk-free rate (r_f)	4.5%					
9		Beta	1.2					
10		Market risk premium (r_{mf})	6.0%					
11								
12		Discount rate (r)	11.7%	=C8+C9*C10				
13		($r = r_f + beta \times r_{mf}$)						